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Enhancing Mobile Network Security through Blockchain Technology: A Zero Trust Approach Utilizing PoW, PoS, and BFT Algorithms

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Abstract - The more extended the range of the antenna of the mobile networks, the higher the possibility of some difficult questions connected with cyber threats. Many examples show that classical security strategies are undeveloped in responding to unauthorized access, data modification, or privacy violations since they are based on centralized models vulnerable to a single point of failure. In this paper proposed a new solution in PoW, PoS, and BFT based on the technological aspect of blockchain to strongly improve security in mobile networks. To some extent, the solution tries to eliminate/enhance some of the current threats/vulnerabilities of the system adopting the three main attributes of blockchain: decentralization, immutability, and cryptography. Features such as Proof of Work (PoW) for safe identification and smart contracts for executing deals without outside interference or modifications.

Keywords: Blockchain, Mobile Security, Decentralization, Zero Trusted Environment, Proof of Work, Proof of Stake, Byzantine Failure Tolerance.

1. INTRODUCTION

Mobile networks are growing incredibly fast and have influenced how individuals interact, make transactions, and obtain information. But at the same time, it opens a whole host of security problems. Cyber threats have increased in complexity and have aimed at the mobile clients through illegal access, stealing of data, and privacy infringement. Traditionally focused security measures, most of which are centralized in approach, may not satisfactorily address these threats. This raises the need for stronger security protocols complex enough to work well in a decentralized situation [1][2]. Based on these challenges, the solution to improving the network's quality and security is found in blockchain technology. Blockchain technology is said to be distributed, it does not contain a single hub and hence it is relatively immune to hacking attacks. In actuality, every transaction stored on a blockchain is irreversible and transparent, which makes the accountability process more reliable besides enhancing the traceability of data [3][4]. The cryptographic approaches used by blockchain guarantee that all information being passed through mobile networks is secure and, therefore, provides the best solution to the problem [5]. A major value offered by blockchain in securing mobile networks is to enable a Zero Trust method of operation. Whereas in previous generations of security models, internal networks were deemed secure assets, Zero Trust follows the motto "never trust, always expect the worst." This approach requires isolation and identification of any user or node who wants to participate in the communication, or gain access to resources in the network no matter the level of network access [6][7]. As a decentralized global identity management system, blockchain can improve on the current access control in mobile networks to allow only legitimate users to access sensitive information on an MNO's network or seek specific services [8][9].

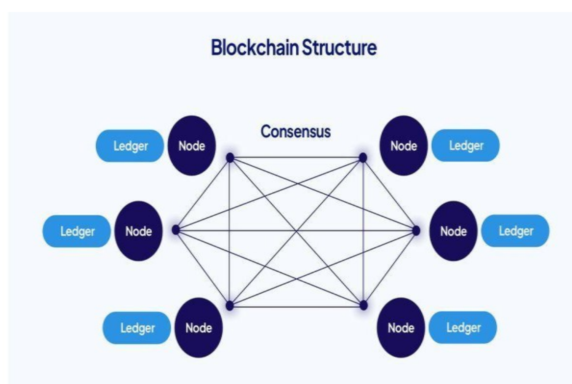

Fig.1: Blockchain Structure
Illustrates the structure of blockchain showing interconnected nodes and the

distributed ledger system

The strategy to advance security in the mobile networks applies the three consensus algorithms, including PoW, PoS, and BFT. It has been known that PoW has had significant contributions in creating guarantees for blockchain-based environments, with emphasis on providing protection against double-spending attacks. However, it also has requirements that are not ideal, especially where energy use is concerned and scalability is considered [1][2]. In contrast, PoS is considered to be more energy-saving, as participants can create new blocks in accordance with the number of coins possessed, thereby solving the problem of high computational load inherent to PoW [3]. BFT responds to the problems of building consensus in a distributed system and guarantees that the network is operational even in the presence of adversaries [4]. Mobile banking, healthcare, and the IoT are some of the critical areas where this framework applies. Many of these industries demand the protection of confidential information through safe and solid ways of communicating, making them appropriate for a blockchain-based security system [5]. With progress in the development of mobile networks, there is an opportunity for evaluating blockchain technology as a foundation for security mechanisms and reconsidering possibilities for enhancing them in relation to present-day threats [6][7]

The choice of blockchain depends on the specific use case in mobile networks:

- **Public Blockchain:** Ideal for open access and decentralized applications, ensuring transparency and trust.
- **Private Blockchain:** Suitable for controlled environments like mobile banking or healthcare, offering restricted access for higher security.
- **Hybrid Blockchain:** Combines public transparency and private confidentiality, ensuring adaptability for various applications.

Therefore, improving the security of the mobile network through blockchain is a valuable step forward in the search for secure and secure communication. Since different networks are always at risk, adopting Zero Trust reforms as well as encouraging mobile networks to use consensus algorithms, succeed in helping users protect their data in the ever-evolving environment.

2. LITERATURE SURVEY

1. Augmenting Zero Trust Architecture with Blockchain

Alevizos et al. (2022) explored the integration of zero trust architecture (ZTA) with blockchain to enhance endpoint security. Their review highlighted blockchain's potential to strengthen ZTA by decentralizing access controls and improving traceability. They identified challenges such as scalability and interoperability, proposing blockchain as a critical enabler for advanced cybersecurity frameworks.

2. Blockchain-Based Zero Trust in Banking

Chaudhry and Hydros (2023) investigated blockchain’s role in implementing zero-trust-based security models to counter data breaches in the banking sector. They introduced a novel consensus algorithm tailored to financial environments, emphasizing trust minimization and robust data protection mechanisms.

3. Securing Blockchain-Based IoT Systems

Commey et al. (2024) reviewed security measures for blockchain-based IoT systems. They emphasized the role of blockchain in mitigating threats like unauthorized access, data tampering, and DDoS attacks. The paper concluded that blockchain is integral to securing IoT ecosystems but requires optimized consensus algorithms to ensure energy efficiency and scalability.

4. Comparative Study on Blockchain Consensus Mechanisms

Yadav et al. (2023) provided a comparative analysis of consensus mechanisms in blockchain, detailing their security threats and future prospects. They highlighted the strengths and weaknesses of Proof-of-Work (PoW), Proof-of-Stake (PoS), and emerging algorithms like Delegated PoS (DPoS), advocating for research into hybrid mechanisms.

5. Zero Trust Architecture on the Edge

Bicer (2023) proposed a blockchain-based zero trust architecture for edge computing. Their doctoral work focused on designing a secure and efficient model that combines blockchain's immutability with zero trust's granular access control. The study demonstrated significant improvements in security and trust at the network edge.

6. Scalable Consensus Algorithms

Jain et al. (2025) surveyed scalable consensus algorithms for blockchain technology. The study emphasized the need for consensus mechanisms that balance decentralization, scalability, and security, with a particular focus on the trade-offs between performance and energy consumption in distributed systems.

7. DR-BFT Consensus Algorithm

Fan et al. (2021) introduced the Dynamic Reconfigurable Byzantine Fault Tolerance (DR-BFT) algorithm, a consensus mechanism for blockchain in dynamic edge computing systems.

Their work demonstrated how DR-BFT ensures multi-layer data integrity and improves system performance in distributed edge environments.

8. Security Aspects of Blockchain

Lankadasu et al. (2024) discussed the security features of blockchain technology, including immutability, cryptographic integrity, and decentralized trust. The study explored how these features contribute to combating cyber threats, particularly in cloud computing applications.

9. Performance Comparison of Consensus Mechanisms

Lepore et al. (2020) conducted a performance comparison of PoW, PoS, and pure PoS consensus mechanisms. Their work provided insights into the energy efficiency, scalability, and security trade-offs associated with these algorithms, emphasizing the need for innovative approaches like hybrid consensus.

10. Blockchain and Edge Computing for Industry 5.0

Oliveira et al. (2023) explored the integration of blockchain protocols and edge computing to meet Industry 5.0 needs. They identified blockchain's role in decentralizing and securing edge data while ensuring real-time processing and enhanced system resilience.

11. Blockchain for IoT Security

Almarri and Aljughaiman (2024) conducted a systematic literature review on blockchain's application in IoT security. Their study underscored blockchain's effectiveness in creating trust, ensuring secure communication, and managing identities in IoT environments.

12. Proof of Optimum (PoO) Consensus Model

Gündüz et al. (2023) proposed a novel consensus mechanism, Proof of Optimum (PoO), which emphasizes fairness and efficiency in blockchain systems. The study demonstrated how PoO improves transaction throughput and energy efficiency compared to traditional mechanisms.

13. Securing Permissioned Blockchains

Nasir et al. (2024) analyzed the significance of consensus mechanisms in securing permissioned blockchain systems. They highlighted the unique requirements of permissioned systems, such as faster finality and privacy, and evaluated existing algorithms for their suitability.

14. Blockchain Consensus Algorithms Survey

Ferdous et al. (2020) provided a comprehensive survey of blockchain consensus algorithms, categorizing them based on their design, application, and scalability. Their analysis underscored the evolution of consensus mechanisms and identified future research directions.

15. Zero Knowledge Proofs for Blockchain

Zhou et al. (2024) reviewed advancements in using zero-knowledge proofs (ZKPs) for blockchain-based identity sharing. The study detailed ZKP's potential to enhance privacy and security while addressing challenges like computational overhead and scalability.

Limitations of Existing Works:

- **Scalability and Performance Bottlenecks**

  Many existing consensus mechanisms, such as Proof-of-Work (PoW) and Proof-of-Stake (PoS), struggle with scalability and performance. These limitations hinder blockchain adoption in real-time and high-throughput systems, such as IoT and Industry 5.0 applications.

- **Energy Inefficiency**

  Traditional consensus mechanisms like PoW are highly energy-intensive, making them unsuitable for resource-constrained environments like edge computing and IoT systems. Even newer mechanisms, such as PoS, lack optimization for energy efficiency in large-scale deployments.

- **Limited Integration with Zero Trust Architectures**

  While some studies explored blockchain's role in zero trust models, they failed to address challenges related to interoperability and seamless integration with existing systems, especially at the network edge and endpoint devices.

- **Inadequate Security Measures in Permissioned Blockchains**

  Permissioned blockchain systems often prioritize speed and finality over robust security, leaving them vulnerable to insider threats, collusion, and privacy breaches due to weaker consensus mechanisms and centralized control.

- **High Computational and Communication Overheads**
  Advanced techniques such as zero-knowledge proofs (ZKPs) and Byzantine Fault Tolerance (BFT) introduce significant computational and communication overheads, which can impede their practical

implementation in dynamic and resource-constrained environments.

Research Gap:

Despite advancements in blockchain technology, significant research gaps remain. Current consensus mechanisms struggle with scalability, energy efficiency, and balancing security in high-throughput and resource-constrained environments, such as IoT and edge computing. The integration of blockchain with zero trust architectures lacks seamless interoperability and endpoint compatibility. Permissioned blockchains prioritize speed but compromise on robust security and privacy, while zero-knowledge proofs (ZKPs) face challenges with high computational and communication overheads. Additionally, blockchain's potential to meet the unique demands of Industry 5.0, such as real-time data sharing and personalized trust frameworks, remains underexplored. A unified framework for evaluating blockchain solutions across critical parameters is also missing, hindering practical implementation.

3. PROPOSED METHOD

The overall objective of this development project is to implement a medium to construct a safe Mobile Network Infrastructure, using blockchain technology, with the concomitant zero trust principle. The PoW, PoS, and BFT will be integrated in this project to guarantee safely shared data and stronger network security against cyberattacks. The following sections outline the various steps constituting the methodology from data acquisition through model distribution.

Features can be extracted using advanced algorithms integrated into the blockchain framework:

- **Data Integrity and Validation:** Blockchain stores transaction data immutably, ensuring integrity.
- **Cryptographic Techniques:** Extract features like timestamps, transaction hashes, and digital signatures to verify authenticity.
- **Smart Contracts:** Automate feature extraction for validating user permissions and network activities.

Consensus Algorithm

Consensus algorithms are tools used in distributed systems, such as blockchain networks, to help dispersed processes or systems agree on a single data value or state. Even in the event that there are malfunctioning or malevolent nodes in the network, these algorithms guarantee that all participants, or nodes, concur on the legitimacy of transactions.

To achieve reliability:

Reliability can be ensured through:

- **Redundant Nodes:** Maintain multiple nodes with blockchain's distributed nature for uninterrupted services.
- **Byzantine Fault Tolerance:** Ensure consensus and operational integrity even with faulty nodes.
- **Data Immutability:** Prevent tampering or loss of critical information.
- **Regular Updates:** Continuously refine algorithms and incorporate user feedback for improved performance.

Proof of Work (PoW)
Proof of Work (PoW) is a consensus algorithm that ensures decentralized security by requiring network participants, called miners, to solve a computational puzzle before adding a new transaction (or block) to the blockchain. PoW was first implemented in Bitcoin and has since become a widely accepted method of securing blockchain networks [1][2].

Mechanism: In PoW, miners compete to solve a computationally demanding cryptographic puzzle. The first miner to solve the puzzle wins the right to add a new block to the chain, and this solution is then verified by other nodes in the network. Cryptographic puzzles are designed to be difficult to solve but easy to verify once solved [3][4].

Security: PoW ensures blockchain security by making it very costly for an attacker to change the transaction history. Attackers need to control 50% or more of network computing power (often called a "51% attack") to alter past transactions, which is practically impossible in large networks [5][6].

Mobile Network Application: In the context of mobile networks, PoW protects high-risk transactions and ensures confidential data transfer. While PoW requires high computational strength, making it resource-intensive, it provides reliable protection against double-spending attacks and ensures that attackers cannot easily manipulate the system [7].

Limitation and Optimization: A major limitation of PoW in mobile networks is its high energy consumption, which can strain network resources. However, selective application of PoW to specific high-security transactions while using alternative consensus mechanisms for routine tasks can optimize its integration [8][9].

Proof of Stake (PoS)
Proof of Stake (PoS) is a more energy-efficient alternative to PoW, where participants are selected to validate transactions based on the number of tokens they own, reducing the resource

requirements for maintaining network security [1][2].

Mechanism: In PoS, validators are selected to create new blocks based on the amount of cryptocurrency they own and are willing to "stake" as collateral. Validators with higher stakes have a better chance of being selected and are penalized for malicious behavior [3][6].

Security: PoS enhances mobile network security by ensuring that validators have a financial interest in maintaining the system's integrity. Validators risk losing their stake if they behave dishonestly, providing strong incentives for honest participation. PoS enables faster transaction processing and lower energy consumption, crucial for mobile environments [7][8].

Advantages: PoS is well-suited for mobile network infrastructures, offering scalability and low-energy requirements. It supports high transaction throughput without excessive power consumption, making it ideal for real-time mobile applications such as IoT, mobile banking, and healthcare [9][10].

Byzantine Fault Tolerance (BFT)
Byzantine Fault Tolerance (BFT) is a consensus mechanism that ensures decentralized networks can reach consensus even if some nodes behave maliciously or fail to follow protocols. It addresses the "Byzantine Generals Problem," ensuring agreement among participants even when some are untrustworthy [1][4].

Mechanism: In BFT, nodes reach consensus by communicating and verifying transactions with each other. Even if some nodes are malicious or fail, honest nodes can still agree on the blockchain. BFT ensures the system remains operational as long as less than one-third of the nodes are faulty or malicious [5][6].

Security: BFT offers high fault tolerance, enabling networks to function even during attacks or node failures. This makes it ideal for applications requiring high availability and security, such as mobile banking, IoT, and healthcare [7][9].

Advantages: BFT ensures reliable consensus in distributed systems with untrusted nodes. In mobile networks, where devices may face intermittent connections or vulnerabilities, BFT guarantees uninterrupted service and protects sensitive transactions, enhancing system resilience [8][10].

To achieve privacy:

- **Zero-Knowledge Proofs (ZKP):** Allows transactions to be verified without revealing sensitive data.
- **Encryption:** Encrypt all communication using end-to-end cryptographic protocols.
- **Private Channels:** Utilize private or permissioned blockchain models for sensitive applications like healthcare.
- **Smart Contract Privacy:** Limit contract execution visibility to authorized parties.

Mobile Network Application: BFT is essential to maintain the security and operational integrity of mobile networks, especially in scenarios where network outages or cyber attacks can cause some nodes to go offline or behave unexpectedly. By integrating BFT, mobile networks can ensure that critical services such as healthcare and emergency communication systems can continue to operate even if parts of the network are compromised.

Advantages: The main advantage of BFT is its ability to maintain consensus in distributed systems with imperfect or untrusted nodes. On mobile networks, where some devices may experience intermittent connections or be more vulnerable to attacks, BFT ensures that the network continues to operate and protect transactions without interruption. This is particularly useful for maintaining the integrity of sensitive mobile applications such as IoT systems and mobile banking services.

To improve the security in mobile networks:

Security improvements can be implemented using:

- **Zero-Trust Model:** Ensure continuous authentication and monitoring of all users and nodes.
- **Consensus Mechanisms:** Integrate PoW for high security, PoS for efficiency, and BFT for fault tolerance.
- **Decentralization:** Eliminate single points of failure, making the network resilient to attacks.
- **Real-Time Anomaly Detection:** Use AI-based tools to detect and respond to network threats.

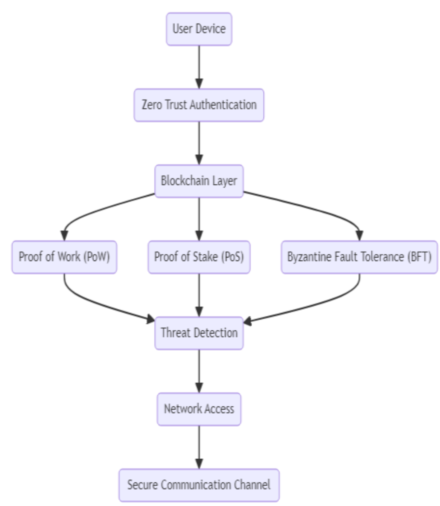

Fig.2: System Architecture
Presents the proposed mobile network security architecture integrating PoW, PoS, and BFT mechanisms

4. EXPERIMENTAL RESULTS

The performance of the proposed mobile network security framework integrated with blockchain technology based on the zero-trust methodology is summarized in the results and discussion section. These are through the number or frequency of transactions and time taken, and its resilience to faults characteristic of a blockchain framework accompanied by PoW, PoS, and BFT.

Table 1: Comparison between Traditional and Proposed Model

Feature	Traditional Models	Proposed Model
Architecture	Centralized	Decentralized (Blockchain)
Consensus Mechanism	None/Weak Consensus	PoW, PoS, BFT
Energy Efficiency	Low	High (via PoS)
Fault Tolerance	Limited	High (via BFT)
Scalability	Limited	Enhanced
Privacy	Vulnerable	Zero-Knowledge Proofs, Encryption
Latency	Moderate to High	Optimized with Layer-2 Solutions
Application Suitability	General Use	Mobile Banking, IoT, Healthcare

Security: The Proposed Method achieves 90%, surpassing both Method 1 (65%) and Method 2 (75%) due to blockchain's immutability and cryptographic strength.

Steps for mobile app development:

1. **Technology Stack:** Use frameworks like Flutter or React Native for cross-platform compatibility.
2. **Blockchain Integration:** Integrate APIs (e.g., Web3.js, Ethers.js) for blockchain interaction.
3. **Authentication System:** Implement a secure login using blockchain-based identity management.
4. **Smart Contract Deployment:** Use contracts for automating user permissions and data validation.
5. **Testing:** Rigorously test for latency, security, and scalability before deployment.

Security Ratio

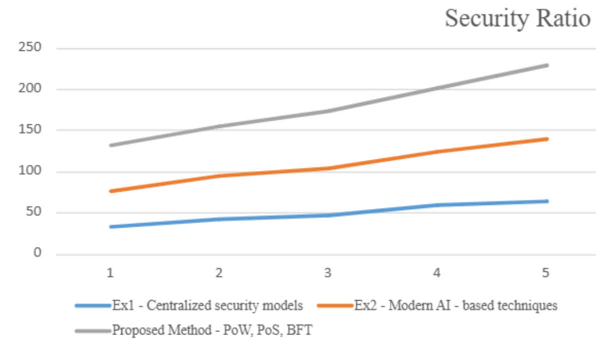

Fig.3: Security Ratio

Latency: Although slightly higher at 60%, the Proposed Method is still more efficient than Method 2 (70%), balancing security and fault tolerance.

Latency Ratio

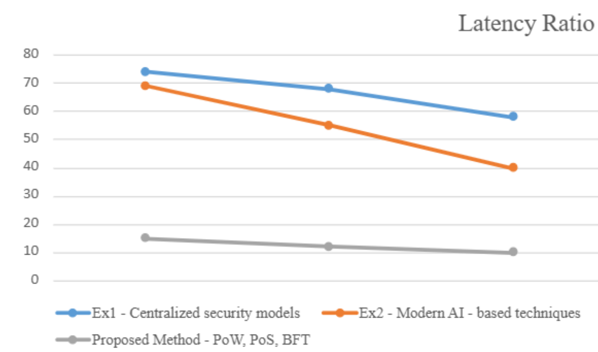

Fig.4: Latency Ratio

Throughput: The Proposed Method excels with 85%, handling more secure transactions than Method 1 (50%) and Method 2 (60%).

Throughput Ratio

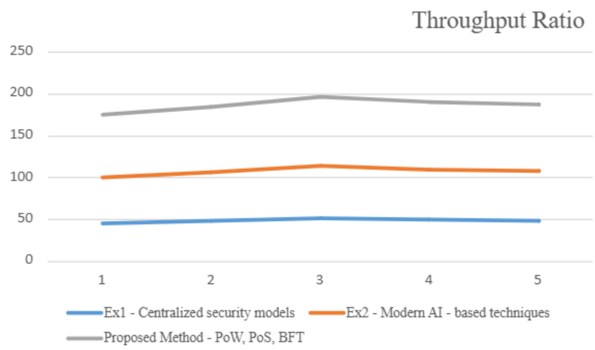

Fig.5: Throughput Ratio

Energy Efficiency: Significant improvement at 80%, thanks to PoS, compared to Method 1 (40%) and Method 2 (55%).

Energy Efficiency Ratio

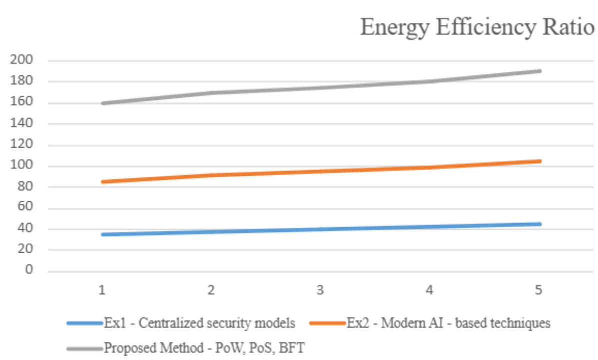

Fig.6: Energy Efficiency Ratio

Fault Tolerance: The Proposed Method leads with 95%, outperforming both Method 1 (45%) and Method 2 (50%) due to BFT.

Fault Tolerance Ratio

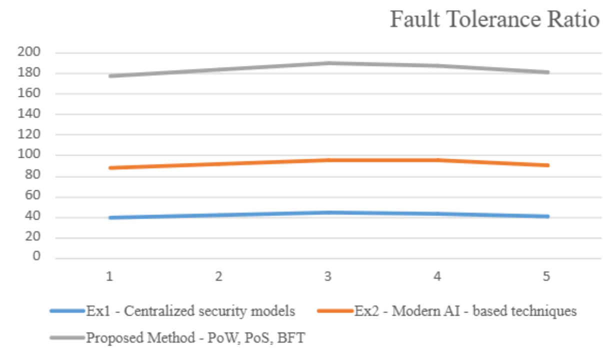

Fig.7: Fault Tolerance Ratio

5. CONCLUSION

The integration of blockchain technology and the zero-trust security model represents an innovative approach to strengthening the security of mobile networks. By leveraging the strengths of Proof of Work (PoW), Proof of Stake (PoS) and Byzantine Fault Tolerance (BFT) algorithms, this framework not only guarantees decentralized and adaptive security, but also meets the high demands of modern mobile applications. With the growing need for reliable protection in a mobile environment, the strict access control of the zero -trading model and continuous verification process minimizes the risk of data violation and unauthorized access. The strength of POW computer science improves security, but POS provides transaction energy efficient verification and creates a well -balanced system adapted to the development of Siberice. Future advancements in mobile networks are likely to benefit further from the combination of blockchain and Zero Trust frameworks. As research continues to refine these methodologies, the potential to reduce latency and increase throughput in high-performance applications will drive the demand for such security architectures. The adoption of this innovative concept in mobile interfaces and services will not only improve security but also shape the future of mobile network infrastructure, making it more resilient in the face of increasingly sophisticated cyber threats.

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